Dynamic Dictionary Learning for Remote Sensing Image Segmentation

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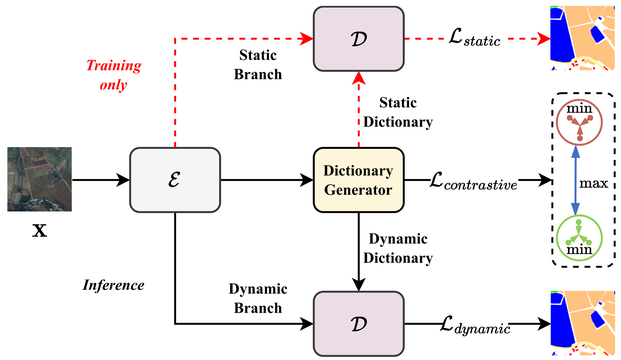
Pipeline



Remote sensing image segmentation struggles with intra-class variation and inter-class similarity. To address this, we introduce dynamic dictionary learning framework for adaptive and discriminative class embeddings.

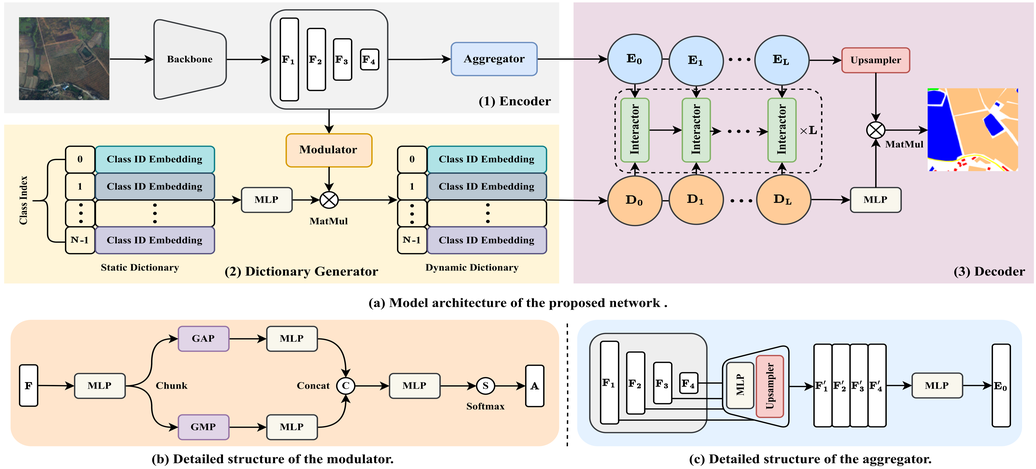
Contributions

- We build dynamic and class-aware dictionaries with contrastive loss for better class separability.
- A modulator converts the static dictionary into a dynamic one, enabling input-specific adaptation.
- A multi-stage alternating cross-attention decoder progressively refines features and embeddings.

Implementation

Our code is publicly available!

Network



Quantitative Results

LoveDA

Method	mIoU (↑, %)
AerialFormer	54.1
SFA-Net	54.9
D2LS (Ours)	**55.3**

UAVid

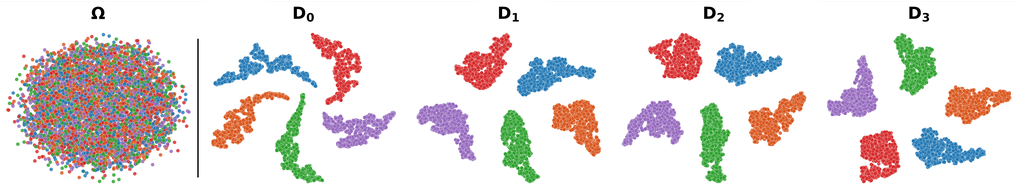
Method	mIoU (↑, %)
UNetFormer	67.8
SFA-Net	70.4
D2LS (Ours)	**70.9**

Potsdam

Method	mF1 (↑, %)
AerialFormer	94.1
SFA-Net	93.5
D2LS (Ours)	**94.7**

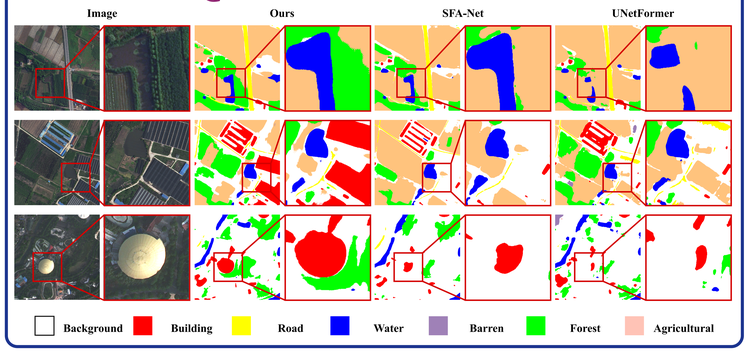
Vaihingen

Method	mF1 (↑, %)
UNetFormer	90.4
SFA-Net	91.2
D2LS (Ours)	**91.9**



Sample Space | Intra-class : 33.61, Inter-class : 94.09 | Intra-class : 28.39, Inter-class : 94.10 | Intra-class : 28.91, Inter-class : 101.78 | Intra-class : 28.51, Inter-class : 102.40

Qualitative Results



Ablation Studies

Effectiveness of Explicit Class Modeling

Dictionary Generator	LoveDA	Vaihingen	Grass
×	54.81	84.40	50.70
√	**55.27**	**85.27**	**51.96**

Static Dictionary vs. Dynamic Dictionary

Loss Function	LoveDA	Vaihingen	Grass
$\mathcal{L}_{\text{static}}$	54.80	85.10	50.96
$\mathcal{L}_{\text{dynamic}}$	54.93	85.17	51.38
$\mathcal{L}_{\text{static}} + \mathcal{L}_{\text{dynamic}}$	54.95	85.17	51.53
$\mathcal{L}_{\text{total}}$	**55.27**	**85.27**	**51.96**